L&T EDUTECH | PCB DESIGN ASSIGNMENT

Buzzer Alert Circuit

with LED Indicator & Push-Button Control

Schematic capture, footprint assignment and PCB layout in KiCad

SUBMITTED BY

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DATE

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TOOL: KiCad | PROJECT: CIA 2

1 Student Details & Project Title

Field	Details
Name of the student	Kumkum Upadhayay
Title	Buzzer Alert Circuit with LED Indicator and Push-Button Control
Software	KiCad (Schematic Editor, Footprint Assignment, PCB Editor, 3D Viewer)
Project file	CIA 2
Date	29-10-2026

2 Project Overview

This project designs a simple alert circuit: when the push button is pressed, a 5 V supply drives an **active buzzer** and, in parallel, a **red LED with a current-limiting resistor**, so the user gets both an audible and a visual indication. The design was taken through the full KiCad flow: schematic, footprint assignment, PCB placement and routing, 2D/3D inspection, BOM and Gerber output.

Circuit working

The 5 V input from connector J2 (through the source V1) reaches the push switch SW1. While SW1 is pressed, current flows into two parallel branches: (a) buzzer BZ1 directly, and (b) resistor R1 in series with LED D1. Both return to GNDPWR. A PWR_FLAG is placed on the ground net so the Electrical Rules Check recognises it as a powered net.

Design calculation (LED branch)

Parameter	Value	Note
Supply voltage	5 V	From J2 / V1
LED forward voltage	about 2 V	Typical for a red LED
R1	330 Ω	As used in the schematic
LED current	(5 - 2) / 330 = about 9 mA	Safe, within the typical 20 mA limit
Power in R1	9 mA x 3 V = about 27 mW	A 1/4 W resistor is more than adequate

Key features

• Push-button controlled alert • Audio + visual indication • Through-hole components for easy soldering
• Standard 2.54 mm / 0.5 mm grid layout

3 Schematic

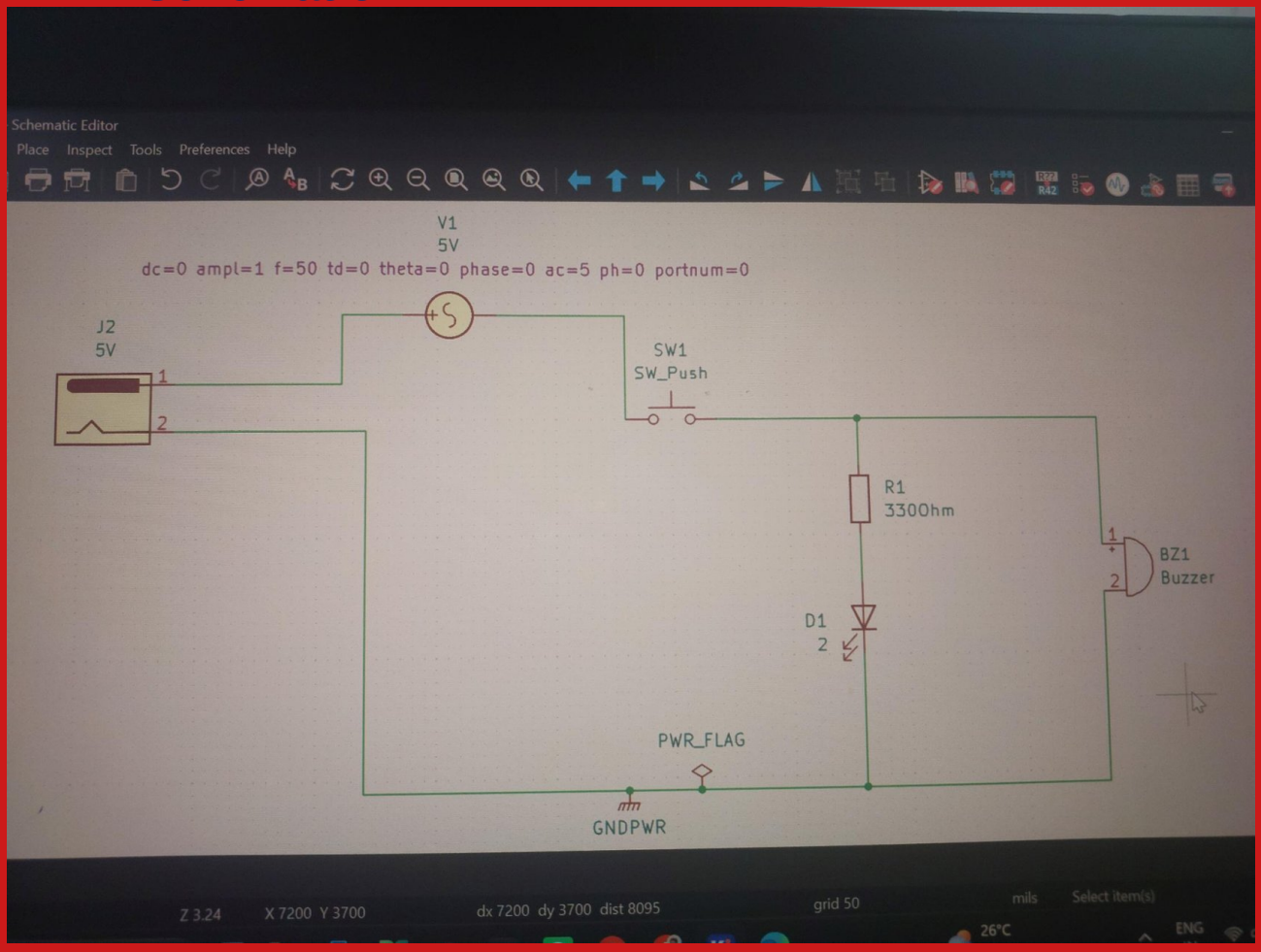


Fig. 1 - Schematic in the KiCad Schematic Editor (J2, V1, SW1, R1, D1, BZ1, PWR_FLAG, GNDPWR)

Ref	Component	Value	Function
J2	2-pin connector	5V	Power input
V1	Voltage source	5V	Supply for simulation
SW1	Push switch	SW_Push	Turns the alert ON while pressed
R1	Resistor	330 Ω	Limits LED current
D1	LED	Red LED	Visual indicator
BZ1	Buzzer	Buzzer	Audible indicator

4 Footprints & PCB Layout

Footprint assignment

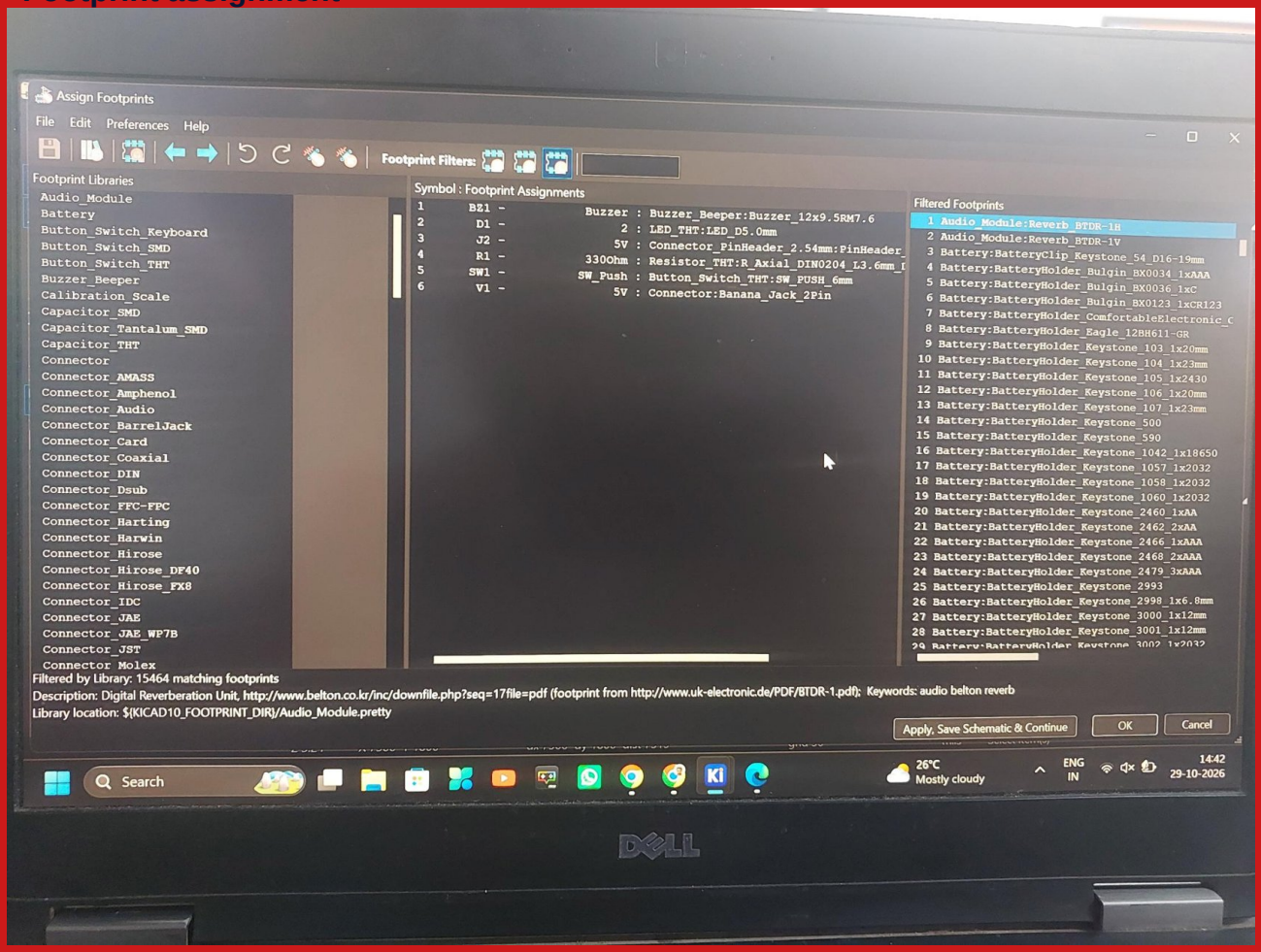


Fig. 2 - KiCad 'Assign Footprints' dialog

Ref	Footprint
BZ1	Buzzer_Beeper:Buzzer_12x9.5RM7.6
D1	LED_THT:LED_D5.0mm
J2	Connector:Banana_Jack_2Pin
R1	Resistor_THT:R_Axial_DIN0204_L3.6mm
SW1	Button_Switch_THT:SW_PUSH_6mm
V1	Connector:Banana_Jack_2Pin

Footprints in the PCB Editor

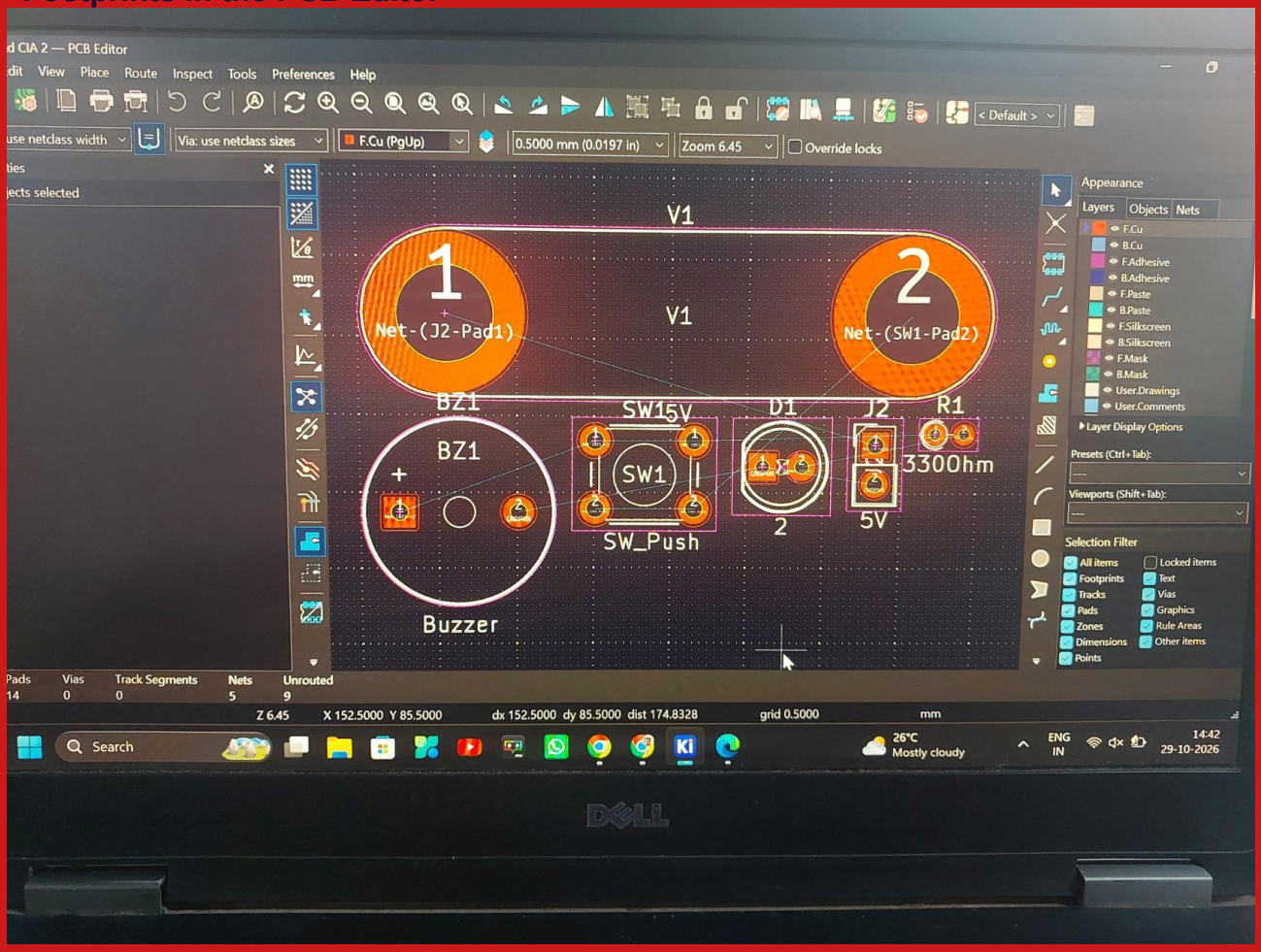


Fig. 3 - PCB Editor: 14 pads, 5 nets. Footprints are placed; ratsnest lines show the connections still to be routed

Note: This screenshot shows the footprints placed with all 9 connections still *unrouted* (Track Segments = 0). Complete the routing on F.Cu with a 0.5 mm track width, then retake this screenshot so it shows the finished tracks, as the assignment requires.

5 2D PCB View

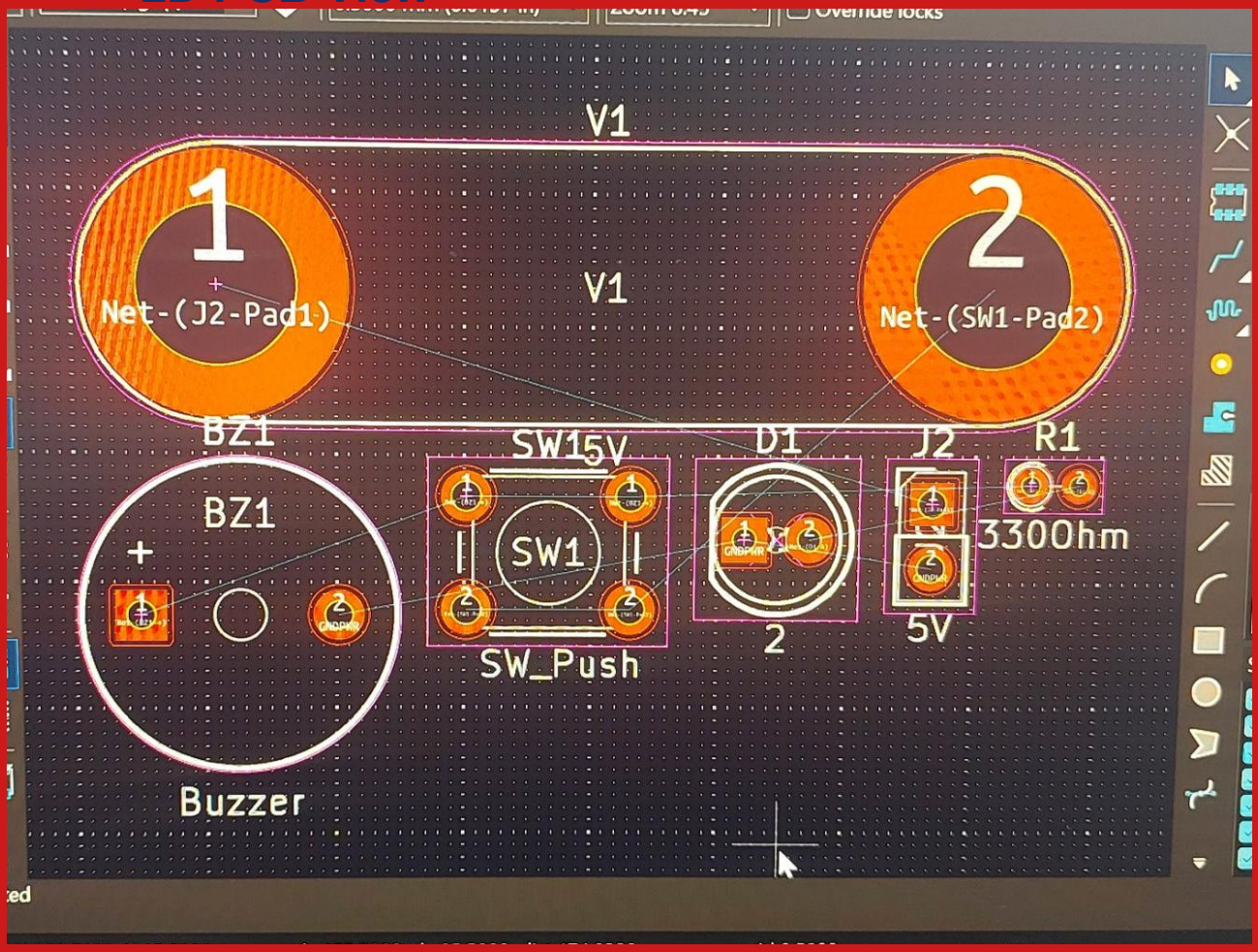


Fig. 4 - 2D view of the board layout (F.Cu layer)

Layout notes

- All components sit on a single side for easy hand-soldering.
- Silkscreen labels (BZ1, SW1, D1, J2, R1) are readable and outside pads.
- The input connector V1 is placed at the top edge, with the switch, LED and resistor grouped to keep tracks short.

6 3D PCB View

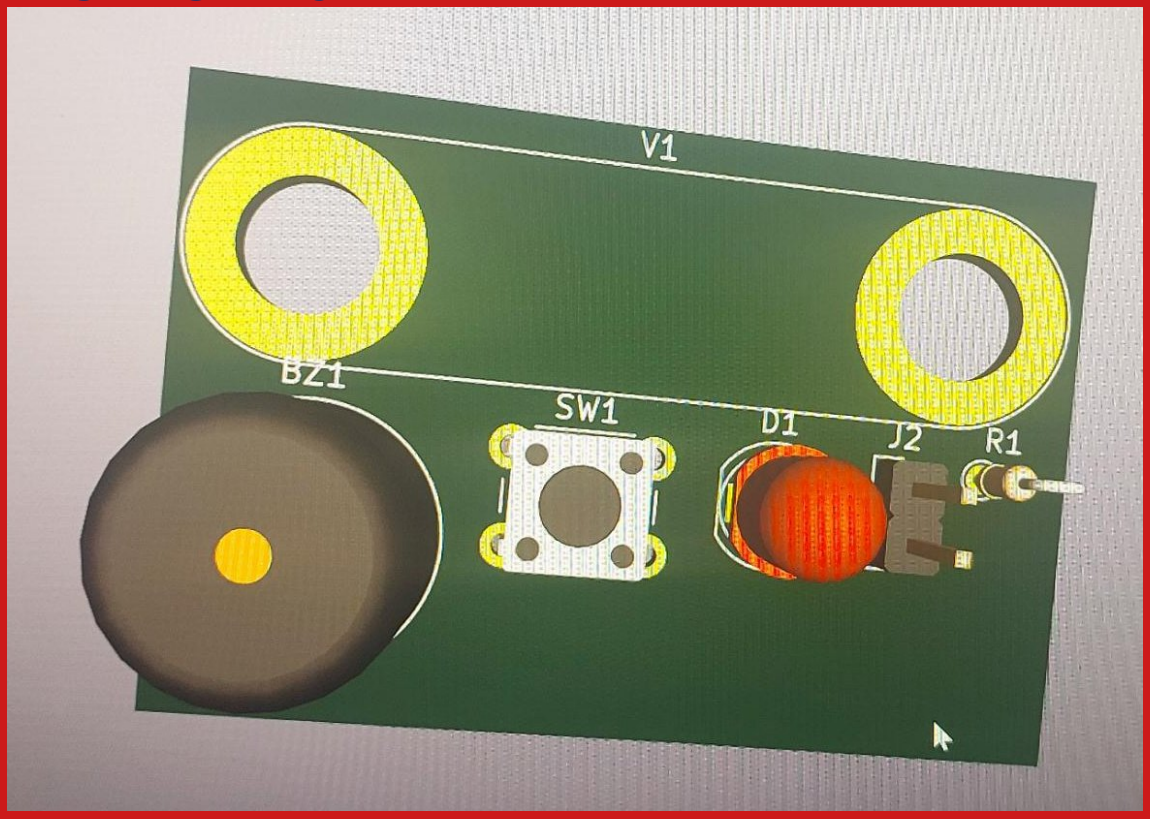


Fig. 5 - 3D Viewer render showing buzzer, switch, LED, connector and resistor models

Warning: KiCad shows 'Board outline is missing or malformed'. Draw a closed rectangle on the **Edge.Cuts** layer around all components and run DRC (Inspect > Design Rules Checker) before exporting Gerbers.

7 Bill of Materials (BOM)

Qty	Ref	Value	Footprint	Description
1	BZ1	Buzzer	Buzzer_12x9.5RM7.6	Active buzzer, 5 V
1	D1	LED	LED_D5.0mm	5 mm red LED
1	J2	5V	Banana_Jack_2Pin	Power input connector
1	R1	330 Ω	R_Axial_DIN0204_L3.6mm	1/4 W carbon film resistor
1	SW1	SW_Push	SW_PUSH_6mm	6 mm tactile push button
1	V1	5V	Banana_Jack_2Pin	5 V source / supply connector

Generate in KiCad: Schematic Editor > Tools > Generate Bill of Materials (export as CSV).

8 Gerber Files

Export steps in the PCB Editor

1. File > Fabrication Outputs > Gerbers (.gbr)
2. Select layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, Edge.Cuts
3. Click **Plot**, then **Generate Drill Files** (Excellon format)
4. Zip the output folder and submit it with this report

Expected file	Layer
*-F_Cu.gbr	Front copper
*-B_Cu.gbr	Back copper
*-F_Silkscreen.gbr	Front silkscreen
*-F_Mask.gbr / *-B_Mask.gbr	Solder masks
*-Edge_Cuts.gbr	Board outline
*.drl	Drill file

Conclusion

The buzzer alert circuit was captured as a schematic, mapped to through-hole footprints and placed in the PCB Editor. With routing, a board outline and a passing DRC, the design is ready for fabrication.